Telecom complaint and service credit assistant

The system in one sentence

Investigate telecom complaints, explain the evidence, and credit eligible outages with clear proof.

Key design calls

Code sets each credit. Rejected: Model-set amounts. Why: Approved rules must give repeatable results.

Start with outage credits. Rejected: All refunds at launch. Why: Speed and bill disputes need staff judgement.

Check uncertain actions first. Rejected: Retry on timeout. Why: A slow reply can hide a posted credit.

Keep a staffed waiting queue. Rejected: Wait inside chat. Why: Cases must survive a two-day wait.

Who acts and who decides

Action	Autonomy
Read the active service and explain records	suggests
Post an eligible outage credit	acts alone
Decide large or unclear adjustments	human decides
Send updates and check uncertain results	acts alone

Top risks by RPN

FM7 | Supervisor queue stalls | RPN 175

FM3 | Duplicate or concurrent credit | RPN 144

FM5 | Unknown commit and cancellation | RPN 144

FM4 | Contradictory or injected evidence | RPN 140

FM9 | Duplicate or wrong-destination notice | RPN 112

Launch blockers and owners

H1: Proposed policy P1: automatic outage... Owner: Billing policy owner.

H2: Account and ledger adapters can... Owner: Integration owner.

H3: Supervisor queue is staffed continuously... Owner: Support operations owner.

H4: Proposed privacy defaults: minimize to... Owner: Privacy owner.

H6: Notice delivery adapter supports stable... Owner: Messaging integration owner.

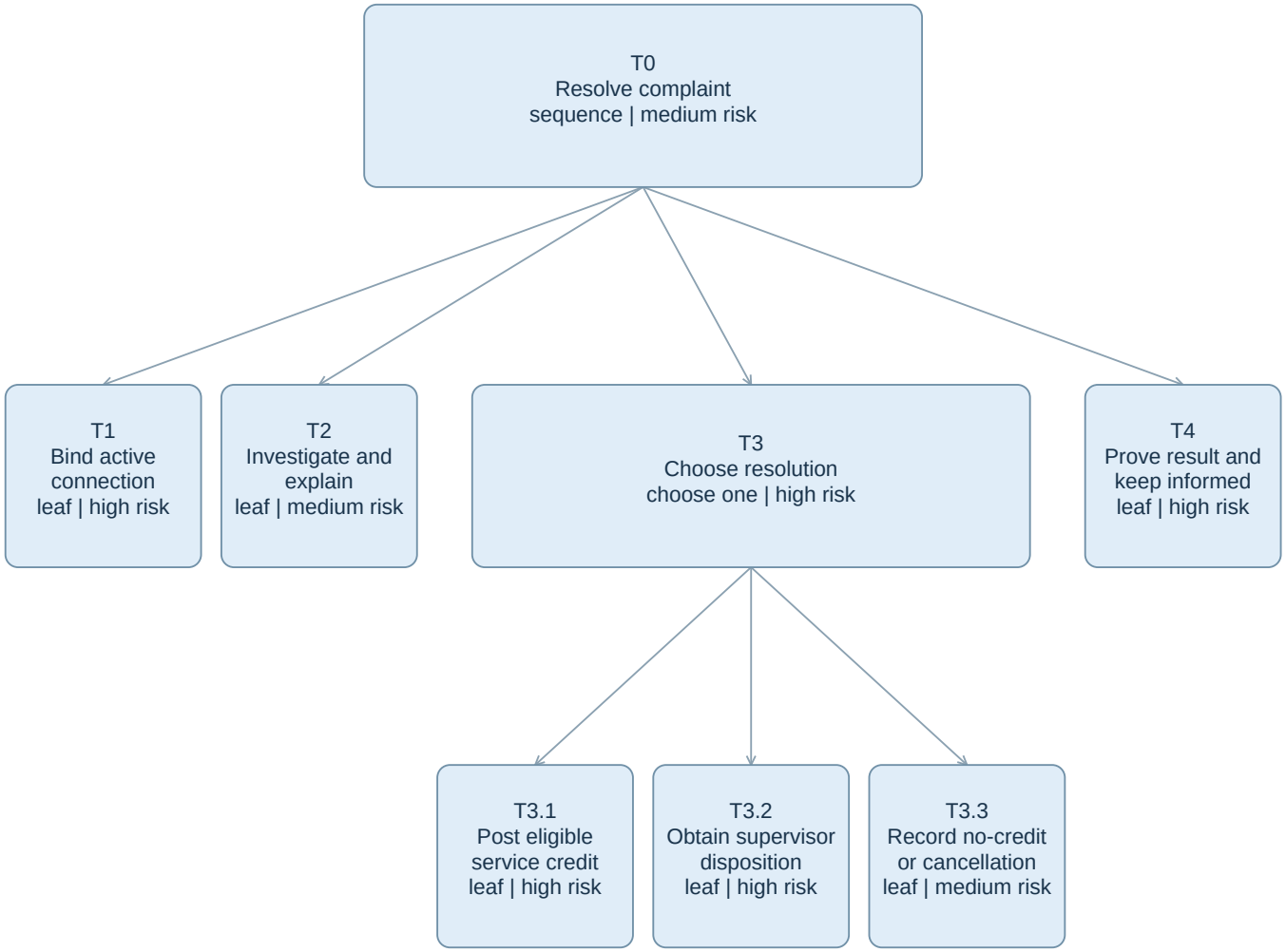
Illustrative cost per case

USD 0.00378 per case in the model budget. This is an estimate, not a current provider quote. Staff and tools cost extra.

Review status: independent-agent; round 1; needs-work.

Proposed design. Full rules and blockers: design.json and assumptions.md.

Task tree



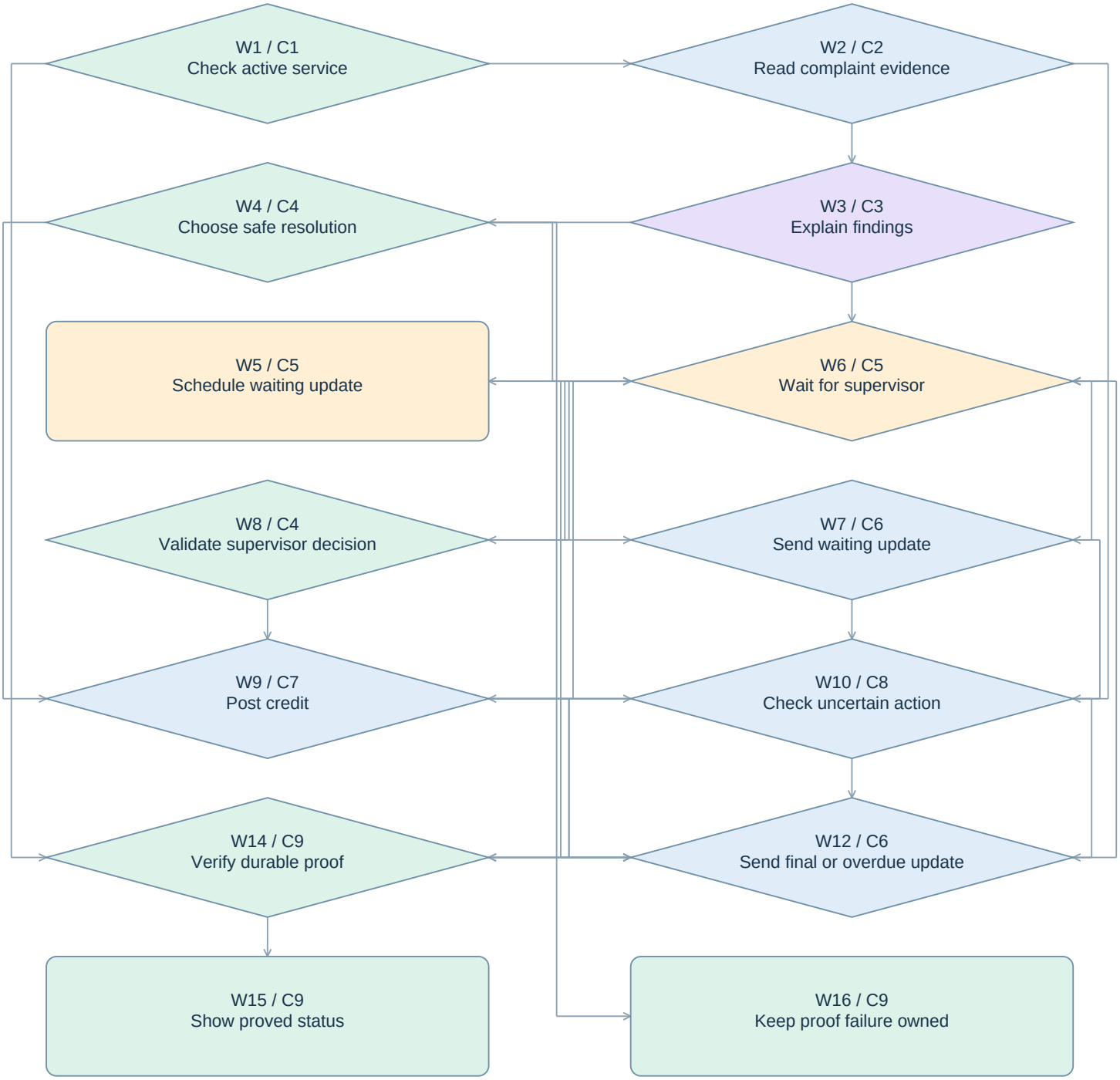
Labels show each leaf mode and risk. Full task inputs, examples and limits are in design.json and PRD.md.

Plain words and scope

First release credits verified outages. Staff investigate speed, billing errors and larger or unclear adjustments.

A fingerprint is a check that the approved details have not changed. A record ID is proof from a tool that the change was saved.

Workflow diagram



Diamonds mark choices. Arrows show all routes, including retry and recovery. Exact branch rules are in architecture.md.

Colour	Role
#E8E0FA	Agents (LLM)
#E0EDF8	Code: tools
#DDF2E9	Code: validation
#FFF0D5	Code: human triggers

Top risks and release checks

Risk heat table - severity x occurrence

Each cell names the risks at that severity and occurrence. The full scores, causes, controls and owners are in fmea-register.csv.

S / O	1	2	3	4	5	6	7	8	9	10
10	-	-	-	-	-	-	-	-	-	-
9	-	-	FM1	FM3, FM5	-	-	-	-	-	-
8	-	-	FM6	FM2, FM8	-	-	-	-	-	-
7	-	-	-	FM9	FM4, FM7	-	-	-	-	-
6	-	-	-	-	-	-	-	-	-	-
5	-	-	-	-	-	-	-	-	-	-
4	-	-	-	-	-	-	-	-	-	-
3	-	-	-	-	-	-	-	-	-	-
2	-	-	-	-	-	-	-	-	-	-
1	-	-	-	-	-	-	-	-	-	-

Risk	Control summary / full register linked
FM1 Wrong customer or inactive line	Gateway identity plus active registry join before read and writer fresh join.
FM2 Stale policy or revoked authority	Re-fetch policy/account/authority versions at writer and require exact immutable command.
FM3 Duplicate or concurrent credit	Atomic incident-cycle unique key plus cycle budget lock; semantic matches get human correlation, no fuzzy automatic second credit.
FM4 Contradictory or injected evidence	Admit scoped fields only; verify citation fields; disagreement routes supervisor.
FM5 Unknown commit and cancellation	Persist operation before submit, lock changed intent, read-only lookup across sessions; no resubmit without final absence.
FM6 False completion	W14 matching posting/case record guard before W15; fallback UNKNOWN_OWNED.
FM7 Supervisor queue stalls	Durable UTC timers, 2h escalation,24h reminder,48h duty-manager disposition; in-app fallback.
FM8 Wrong arithmetic or unsupported adjustment	Integer arithmetic; union intervals, exact cycle/proration; above-cap human route; no automatic speed/billing amount.
FM9 Duplicate or wrong-destination notice	Verified channel, minimized templates, stable notice key and readback after timeout.

Risk scores are proposed estimates. They are not measured odds or risk acceptance.

Components and contracts

This inventory is a summary. Exact typed inputs, outputs and permission rules are in architecture.md and the build milestone files.

Component / group	Responsibility	Task IDs
C1 / Code: validation	Identity and scoped intake	T1
C2 / Code: tools	Scoped evidence reader	T2
C3 / Agents (LLM)	Evidence-grounded explanation	T2
C4 / Code: validation	Rule and action validation	T3.1, T3.2, T3.3
C5 / Code: human triggers	Durable supervisor queue	T3.2, T4
C6 / Code: tools	Customer status delivery	T4
C7 / Code: tools	Service-ledger credit writer	T3.1
C8 / Code: tools	Read-only reconciliation	T3.1, T4
C9 / Code: validation	Record proof and close	T3.3, T4

Shared control boundary

Code checks identity, money, dates and limits. A model may suggest a choice. It cannot authorise or perform a business write.

Architecture and model budget

F1 - Durable code workflow

Transactional case state machine with bounded explanation call; SQL outbox/event log and supervised queue. Rejected: ['Single open-ended tool agent; synchronous chat-only chain.']

Why: Keeps authority explicit and failures owned.

F2 - Action scopes

Separate read, credit and notice service principals; writer fetches exact immutable command and rechecks authority. Rejected: ['Shared administrator token.']

Why: Keeps authority explicit and failures owned.

F3 - Authoritative storage

Relational case/operation ledger and immutable evidence snapshots; no vector store is authoritative. Rejected: ['Put all billing/account state in retrieval index.']

Why: Keeps authority explicit and failures owned.

Model cost

Illustrative model cost is USD 0.00378 per case. Unknown prices are not free service. Confirm rates, traffic, staff and tools before launch.

Full call counts, token budgets, rejected choices and costs: design.json, architecture.md and assumptions.md.

Evaluation coverage matrix

Risk / task	EV1	EV2	EV3	EV4	EV5	EV6	EV7	EV8	EV9	EV10	EV11	EV12
FM1	X	X	-	-	-	-	-	-	-	-	-	-
FM2	X	-	X	-	-	-	-	-	-	-	-	-
FM3	X	-	-	X	-	-	-	-	-	-	-	-
FM4	-	-	-	-	X	-	-	-	-	-	-	-
FM5	-	-	-	-	-	X	-	-	-	-	-	-
FM6	X	-	-	-	-	X	X	-	-	-	-	-
FM7	-	-	-	-	-	-	-	X	-	-	X	-
FM8	X	-	-	-	-	-	-	-	X	-	-	X
FM9	-	-	-	-	-	-	-	-	-	X	X	-
T1	X	X	-	-	-	-	-	-	-	-	-	-
T3.1	X	X	X	X	-	X	-	-	X	-	-	X
T3.2	-	-	X	-	X	-	-	X	-	-	-	X
T4	X	-	-	-	-	X	X	X	-	X	X	-

Evaluation coverage matrix - continued

Risk / task	EV13	EV14	EV15	EV16	EV17	EV18	EV19	EV20
FM1	-	-	-	-	-	X	-	-
FM2	X	X	-	-	-	-	-	-
FM3	-	-	-	-	-	-	-	-
FM4	-	-	-	-	-	-	X	X
FM5	-	-	-	X	X	-	-	-
FM6	-	-	-	-	-	-	-	-
FM7	-	-	-	-	-	-	-	-
FM8	X	-	-	-	-	-	X	-
FM9	-	-	X	-	-	-	-	-
T1	-	-	-	-	-	X	-	-
T3.1	X	X	-	X	X	X	-	-
T3.2	X	X	-	-	-	-	X	-
T4	-	-	X	X	-	-	-	-

X means the test is linked to that risk or task. It does not mean the test passed. Complete fixtures and assertions: eval-set.json.

Evaluation plan

Test	Layer	Purpose
EV1	golden	Valid outage arithmetic and proof
EV2	golden	Cross-customer service rejected
EV3	golden	Policy changed after decision
EV4	golden	Two operations same outage serialize
EV5	golden	Conflicting outage cannot authorize text instruction
EV6	golden	Unknown commit after logout and cancellation
EV7	golden	No proof means no success
EV8	golden	48-hour queue retains owner and visible status
EV9	golden	Missing rule blocks calculation
EV10	golden	Wrong notification destination rejected
EV11	golden	Verified waiting notice delivered
EV12	golden	Large calculated credit goes to human
EV13	golden	Valid supervisor-bound credit
EV14	golden	Revocation wins after approval
EV15	golden	Notice timeout readback prevents duplicate send
EV16	golden	Unknown stays locked at polling bound
EV17	golden	Cancellation before submit suppresses credit
EV18	golden	Inactive connection cannot receive credit
EV19	golden	Poor speed investigated but not auto-credited
EV20	golden	Invented citation triggers template fallback
EV21	offline	Stratified end-to-end evidence review
EV22	online	Operational safety monitors
EV23	release	Release suite and approval evidence

These are test plans. No business controls, provider quality or user outcomes were measured. See eval-set.json for fixtures and pass rules.

Measurement and rollout

R1 - Measure safe resolution

Resolution hours: median/P90 from complaint intake to proved outcome; unresolved remains censored and separately aged. Correctly resolved/all eligible cases over 7 days, duplicates and wrong-subject credits per 1000 attempts. Baseline unknown: two weeks observation. Proposed improvement target 20%...

Why: Keeps authority explicit and failures owned.

R2 - Shadow to assist to pilot to scale

Shadow entry: sandbox credentials and approved test data; exit: EV1-EV23 evidence without business writes. Assist entry: H1-H4,H6 verified and staff trained; exit: two weeks representative expert-reviewed decisions meeting EV21, zero safety violations. Pilot entry: assist evidence and signed...

Why: Keeps authority explicit and failures owned.

R3 - Account-randomized controlled pilot

Collect two-week baseline for binary safe-resolution rate and log-duration variance. Pre-register minimum detectable improvement, alpha .05, power .8, baseline proportions for binary calculation; for log-hours use two-mean variance formula, cluster/design effect from repeated cases per account. No sample...

Why: Keeps authority explicit and failures owned.

R4 - Operating ownership and launch evidence

Billing policy owner approves H1 and exception authority. Integration owner proves H2/H6 adapters and incident recovery. Support operations approves H3 staffing/timers. Privacy owner signs H4. Product owner approves H5 budget and expansion. Retain evidence links; no actual approvals...

Why: Keeps authority explicit and failures owned.

Build handoff and review

Start with build/00-overview.md. Load one milestone file at a time. Exact source rules remain in design.json; do not treat design approval as live write permission.

X1: Coding handoff preserves the first-release credit boundary, exact authority and recovery contracts, and launch evidence gates.

Review dimensions - evidence in review.json

Dimension	Score	Dimension	Score
D1	3/4	D7	2/4
D2	3/4	D8	3/4
D3	3/4	D9	3/4
D4	3/4	D10	2/4
D5	3/4	D11	3/4
D6	3/4	D12	3/4

IR1 major OPEN: D10 lacks substantive boundary and adversarial fixtures for money/authority controls: no automatic-cap equality/overflow pair, remaining cumulative-charge budget case, overlapping outage interval union case, approved...

IR2 minor OPEN: S8.5: C1 accepts optional selection and rejects wrong/inactive selections, but does not explicitly require clarification when multiple active connections exist and none is selected...

IR3 major OPEN: D7 comparison is deferred rather than performed: Temporal and LangGraph are listed for a future spike, and AWS/GCP/Azure descriptions are identical generic placeholders. Provide...

IR5 observation OPEN: semantic-contract.md describes actual golden input/expected as JSON objects while stage-shaped agreed EV1-EV20/eval-set store serialized JSON and semantics stores parsed object copies. The bundled validator...

Glossary

canonical: agreed

discriminated: typed

payload binding: approved details

digest: fingerprint

idempotency: duplicate-write protection

reciprocal: two-way

terminal disposition: final outcome

calc_digest: a fingerprint of the approved amount

After delivering DESIGN.pdf, ask the user: Would you also like a visual HTML version in the same ten-stage structure as the app? Generate HTML only after opt-in, unless already requested.

Full assumptions, owners and evidence: assumptions.md. Review method and failed rounds are preserved in review.json and review-round-*.json.